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Design and Implementation of a Fitness Application Based on HarmonyOS

Xiaoju Ma[1,a*]
[1]School of Information Engineering,
Xi'an Eurasia University,
Xi'an, China
[a*]Corresponding author: maxiaoju@eurasia.edu

Cong Chen[2,b]
[2]School of Information Engineering,
Xi'an Eurasia University,
Xi'an, China
[b]chencong1@eurasia.edu

***Abstract*—With the rapid development of mobile technology, fitness applications based on intelligent operating systems have become an important tool for health management. This article designs and implements a fitness app based on HarmonyOS to provide users with personalized activity records, training plans, and data analysis functions. The system adopts object-oriented system analysis and design methods, and its main functions include user management, sports data recording, training plan formulation, social interaction, etc. The database is designed with a relational model to support the storage and management of user information, exercise records, training plans and other data. This article also discusses the application potential of HarmonyOS in the field of mobile health, and analyzes the technical implementation details of the system, including the development environment, technology stack, and implementation of key functional modules. Through system testing, its functional integrity and performance stability were verified. This study provides a reference for the development of fitness applications based on HarmonyOS, and provides theoretical support for further optimization of mobile health services.**

Keywords- HarmonyOS, Fitness App, Mobile Application, Database Design

I. INTRODUCTION

With the rise of global health awareness, fitness and health management are gradually becoming an important part of daily life. The rapid development of mobile technology has provided new solutions for personal health management, and the popularity of smartphones and wearable devices has made fitness apps an important tool for recording exercise data, developing training programs, and monitoring health conditions[1]. Most of the existing fitness apps rely on traditional mobile operating systems. Fitness apps based on the Android and iOS platforms (e.g., MyFitnessPal and Strava) have integrated functions such as exercise data recording, diet management, and social interactions to provide users with comprehensive health management services[2-3]. However, these apps still fall short in terms of cross-device collaboration and data privacy protection[4-6].

Huawei's HarmonyOS provides a new technology platform for fitness app development with its distributed architecture, efficient performance, and cross-device collaboration capabilities. The OS supports multiple device types such as smartphones, tablets, smartwatches and IoT devices. Its distributed data management and task scheduling capabilities enable fitness applications to seamlessly connect multiple devices for real-time data synchronization and efficient processing.

This study aims to design and implement a HarmonyOS-based fitness application to provide users with personalized fitness management services through systematic functional design and database optimization. This paper analyzes the functional requirements of the fitness application, describes the core functional modules through use case diagrams, designs an efficient database model to support the storage and management of data such as user information, exercise records, and training plans, and explores the technical advantages of HarmonyOS in the development of fitness applications, and proposes future research directions.

II. SYSTEM REQUIREMENTS ANALYSIS

The system has three roles: administrator, user, coach. Ordinary users can use the APP to make fitness records, view data and so on. Professional fitness trainers, can create training plans, view trainee data, etc. System administrator, responsible for user management, data maintenance and so on. The system requirements model is shown in Figure 1.

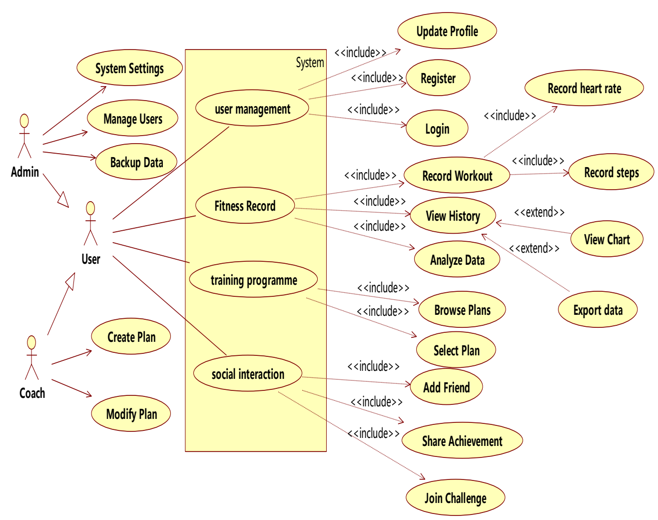

Figure 1. Use Case Diagram.

III. SYSTEM DESIGN

A. System Architecture Design

This system adopts a layered architecture design, mainly including the following four layers:

Presentation Layer: Responsible for interacting with users and providing a friendly user interface. It is developed based on

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the UI framework of HarmonyOS and supports multiple device types, such as cell phones, tablets and smart watches.

Business Logic Layer: Responsible for core business logic, such as exercise data recording, training program management and data analysis. Developed in Java or JavaScript, it runs in the HarmonyOS runtime environment.

Data Access Layer: Responsible for interacting with the database to provide persistent data storage and access functions. SQLite database is used to store structured data, such as user information, exercise records and training programs.

Data Storage Layer: Responsible for storing user data, including local storage and cloud storage. Local storage is used to store user's private data and offline data, while cloud storage is used to backup data and synchronize data across multiple devices.

B． Functional Module Design

The system mainly contains 6 functional modules, as shown in Figure 2 :

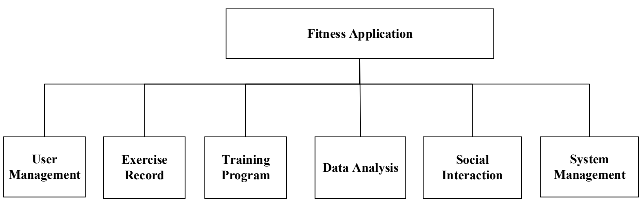

Figure 2. System Functional Modules.

User management module: responsible for user registration, login, personal information management and other functions.

Exercise Record Module: Responsible for recording the user's exercise data, such as steps, distance, calorie consumption, heart rate, etc. It supports both manual input and automatic collection. It supports manual input and automatic collection.

Training Program Module: Provides functions of browsing, selecting, creating and modifying training programs. Users can choose suitable training programs according to their own conditions, and coaches can create and release training programs.

Data Analysis Module: Analyzes the user's exercise data and generates charts and reports to help the user understand his/her exercise situation and progress trend.

Social Interaction Module: Provides functions such as adding friends, sharing exercise results, and participating in challenges to enhance user interaction and fitness motivation.

System Management Module: Responsible for user management, data backup, system settings and other functions, open to administrators only.

C． Database Design

1) Design Principles

This system design is based on the HarmonyOS platform and adopts a layered architecture and modular design, providing functional modules such as user management, exercise records, training plans, data analysis and social interaction. The database design adopts a relational model, which can effectively store and manage user data. The system design has good security, performance and scalability, and provides a reference for the development of fitness applications based on HarmonyOS.

2) Conceptual Model

The system consists of four entities: User, PlanDetail， WorkoutRecord and TrainingPlan. The relationship between entities is shown in Figure 3.

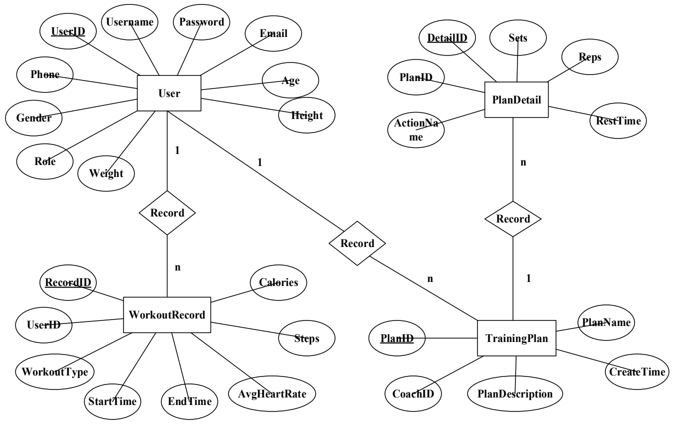

Figure 3. E-R Diagram.

3) Data Table Design

The system mainly contains five tables 1- 5: User, PlanDetail, WorkoutRecord and TrainingPlan,Friend.

Table 1 TrainingPlan

id	FieldName	DataType	PK?	Not NULL	Comment
1	PlanID	INT	Y	N	Primary Key
2	CoachID	INT	N	Y	Primary Key
3	PlanName	VARCHAR(100)	N	Y	
4	PlanDescription	TEXT	N	Y	
5	CreateTime	DATETIME	N	Y	

Table 2 user

id	FieldName	DataType	PK?	Not NULL	Comment
1	UserID	INT	Y	N	Primary Key
2	Username	VARCHAR(50)	N	Y	
3	Password	VARCHAR(50)	N	Y	
4	Email	VARCHAR(100)	N	Y	
5	Phone	VARCHAR(20)	N	Y	
6	Gender	VARCHAR(1)	N	Y	
7	Age	INT	N	Y	
8	Height	FLOAT	N	Y	
9	Weight	FLOAT	N	Y	
10	Role	VARCHAR(20)	N	Y	

Table 3 WorkoutRecord

id	FieldName	DataType	PK?	Not NULL	Comment
1	RecordID	INT	Y	N	Primary Key
2	UserID	INT	N	Y	Foreign Key

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3	WorkoutType	VARCHAR (50)	N	Y	
4	StartTime	DATETIME	N	Y	
5	EndTime	DATETIME	N	Y	
6	Calories	FLOAT	N	Y	
7	Steps	INT	N	Y	
8	AvgHeartRate	FLOAT	N	Y	

Table 4 PlanDetail

id	FieldName	DataType	PK?	Not NULL	Comment
1	DetailID	INT	Y	N	Primary Key
2	PlanID	INT	N	Y	Foreign Key
3	ActionName	VARCHAR (100)	N	Y	
4	Sets	INT	N	Y	
5	Reps	INT	N	Y	
6	RestTime	INT	N	Y	

Table 5 Friend

id	FieldName	DataType	PK?	Not NULL	Comment
1	FriendID	INT	Y	N	Primary Key
2	UserID1	INT	N	Y	Foreign Key
3	UserID2	INT	N	Y	Foreign Key
4	Status	VARCHAR (20)	N	Y	

IV. SYSTEM IMPLEMENTATION

Based on the HarmonyOS platform, this system is developed using Java language to realize the functions of user login, exercise data recording, and training plan browsing. The key code and implementation details show how to use the APIs and components provided by HarmonyOS for application development. With reasonable code structure and modular design, the system has good maintainability and scalability.

A. Development Environment

In Operating System: Windows 10/11 or macOS

Development Tools: DevEco Studio (official HarmonyOS IDE)

Simulator: HarmonyOS Simulator (for testing and debugging)

Versioning: Git (for code versioning).

B. Technology Stack

Programming Languages: Java (primary), JavaScript (optional)

UI Framework: Java UI Framework (HarmonyOS-based UI components)

Database: SQLite (embedded relational database)

Network communication: HTTP/HTTPS, WebSocket (for communicating with cloud servers)

Data format: JSON (for data exchange).

C. Key Code Implementation

1) User Login Function

Users need to agree to the agreement when logging in to the app for the first time, as shown in Figure 4. When the user agrees to the agreement, the user is taken to the app task addition page, as shown in Figure 5.

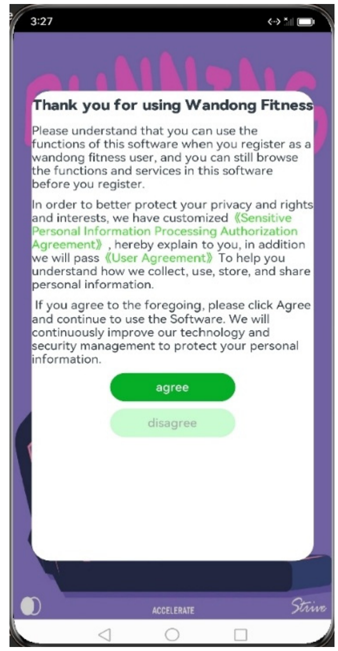

Figure 4. Homepage.

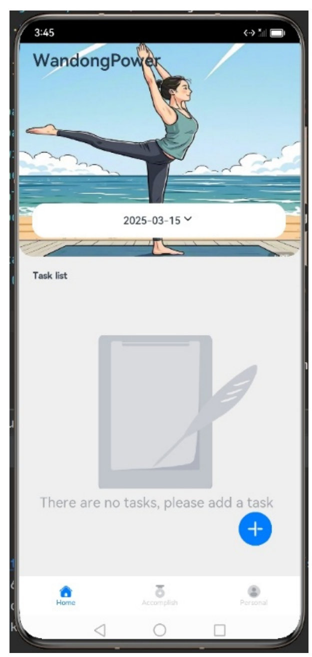

Figure 5. TastList.

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The user can add a training task by clicking the Add button, as shown in Figure 6.

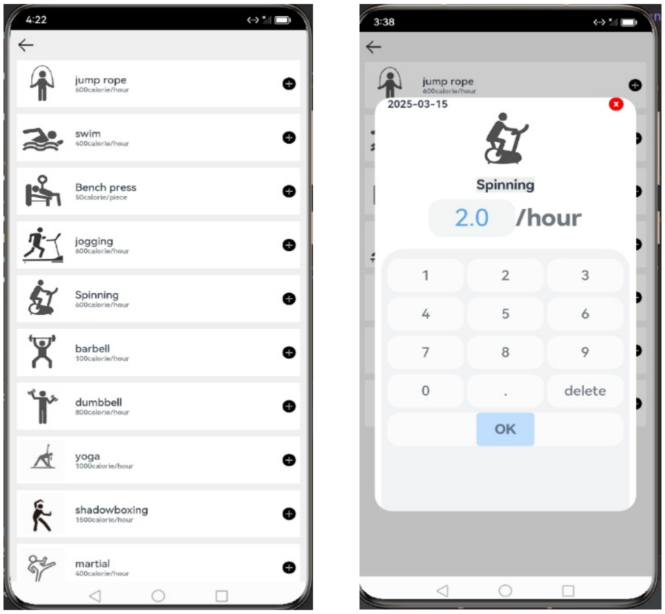

Figure 6. addTast.

2) *Motion data recording function*

After completing a task in the task list, users can view the completion status of their tasks and obtain corresponding badges according to the number of days they have completed, as shown in Figure 7.

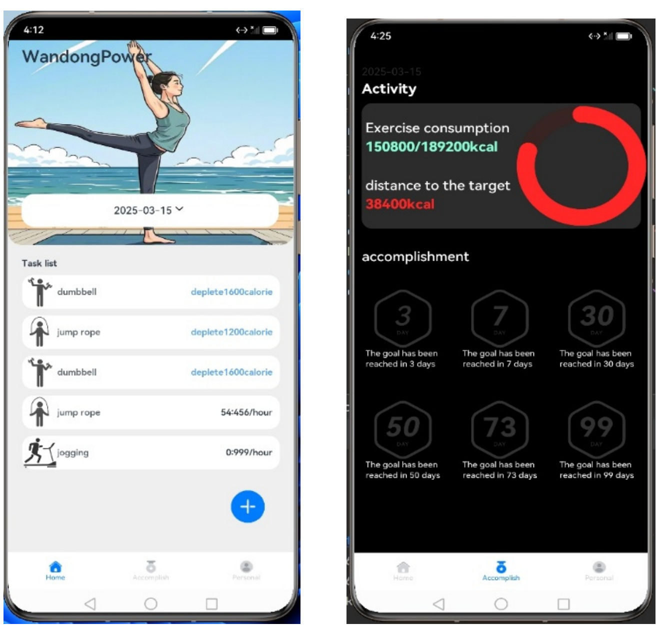

Figure 7. accomplishment page.

V. SYSTEM TESTING

This system uses a combination of black-box testing and white-box testing to comprehensively test the system's functionality, performance, security and other aspects.

A. *Test cases*

1) *Functional Testing*

The functional test was mainly tested for four modules: user registration, login, exercise recording, and training plan.Table 6 lists the user registration function test, and Table 7 lists the user login function test. The functional test of sports data recording is shown in Table 8. Table 9 shows the functional tests of the training plan.

Table 6 Test Case 1: User Registration

Test Item	Test Data	Expected Results	Actual Result	Test Results
username	null	'User name cannot be empty'.	'User name cannot be empty'.	pass
username	Existing user name	'User name already exists'.	'User name already exists'.	pass
username	Compliant Username	Successful registration	Successful registration	pass
password	null	'Password cannot be empty.'	'Password cannot be empty.'	pass
password	Length less than 6 digits	'Password length cannot be less than 6 digits.'	'Password length cannot be less than 6 digits.'	pass
password	Compliant password	Successful registration	Successful registration	pass

Table 7 Test Case 2: User Login

Test Item	Test Data	Expected Results	Actual Result	Test Results
username	null	'User name cannot be empty'.	'User name cannot be empty'.	pass
username	Non-existent username	'User name or password error'.	'User name or password error'.	pass
username	Correct Username	Login Successful	Login Successful	pass
password	null	'Password cannot be empty.'	'Password cannot be empty.'	pass
password	Wrong password	'User name or password error'.	'User name or password error'.	pass
password	Right password.	Login Successful	Login Successful	pass

Table 8 Test Case 3: Motion Data Logging

Test Item	Test Data	Expected Results	Actual Result	Test Results
Start time	current time	Record the start time	Record the start time	pass
end time	current time	Record the end time	Record the end time	pass
Type of sport	jogging	Record the type of exercise	Record the type of exercise	pass
Step count	1000	Record steps	Record steps	pass
calorie	200	Record calories	Record calories	pass
heart rate	80	Record heart rate	Record heart rate	pass

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Table 9 Test Case 4: Training Programme View

Test Item	Test Data	Expected Results	Actual Result	Test Results
A list of training plans	Get it from the database	Displays a list of training plans	Displays a list of training plans	pass
Training program details	Click a training plan	Displays the details of the training plan	Displays the details of the training plan	pass

2) Performance Testing

The system performance test mainly tests the system running time, concurrent processing of dynamic data records, and SQL injection, as shown in Table 10-12.

Table 10 Test Case 1: User Login Response Time

Test Item	Test Data	Expected Results	Actual Result	Test Results
User login	100 concurre nt users	The average response time is less than 1 second	The average response time is 0.8 seconds	pass

Table 11 Test Case 2: Concurrent Processing of Motion Data Logging

Test Item	Test Data	Expected Results	Actual Result	Test Results
Movem ent data recordi ng	100 concurrent requests	All requests were successfully processed	All requests were successfully processed	pass

3) Security Testing

Table 12 Test Case 1: SQL Injection Attack

Test Item	Test Data	Expected Results	Actual Result	Test Results
User login	Username: ' OR '1'='1	Login failed	Login failed	pass

The functional integrity, performance stability and security of the system were verified through system testing. The test results show that the system can meet the user requirements and has good performance and security.

VI. CONCLUSIONS AND PROSPECTS

This study designs and implements a HarmonyOS-based fitness application that aims to provide users with personalised fitness management services. Through systematic functional design, database optimisation and performance testing, we have successfully developed a well-functioning, stable and safe and reliable fitness application. The following are the main results and contributions of this research:

A. Research Achievements

1) System design and implementation

Based on the HarmonyOS platform, adopting layered architecture and modular design, we have implemented core functional modules such as user management, exercise data recording, training plan development, data analysis and social interaction. The system has good scalability and maintainability.

2) Database Optimisation

An efficient database model is designed to support the storage and management of data such as user information, exercise records and training plans. The read and write performance of the database is improved by index optimisation and query statement optimisation.

3) Performance and Security

The stability and security of the system in high concurrency scenarios are verified through performance and security tests. The system can effectively defend against common network attacks, such as SQL injection and XSS attacks.

B. Future Work Directions

Although this research has achieved certain results, there are still some aspects that need to be further explored and improved:

1) Application of Artificial Intelligence

Introducing AI techniques, such as machine learning algorithms, to analyse and predict the user's exercise data and provide more personalised fitness advice.

2) Multi-device optimisation

Optimise the distributed capabilities of HarmonyOS to achieve seamless collaboration between mobile phones, smartwatches, smart home devices and other devices to enhance the user experience.

3) Health Data Privacy Protection

Research on more advanced encryption technology and privacy protection solutions to ensure the security and privacy of user health data.

4) User Behaviour Analysis

Analyse user behaviour data to understand user needs and preferences, optimise system functions and services, and improve user satisfaction.

In conclusion, this study provides theoretical support and practical reference for the development of fitness applications based on HarmonyOS, and promotes the development of mobile health technology. In the future, we will continue to explore new technologies and methods to provide users with more intelligent, convenient and safe fitness management services.

REFERENCES

[1] Yuan Jiangchen. Design and implementation of fitness reservation management system[J]. Fujian Computer,2023,39(11):111-113.

[2] LIU Zhuoxin,FENG Yuechun. Design and implementation of AI personal fitness trainer system based on LLM and CV[J]. Henan Science and Technology,2025,52(02):24-28.

[3] Seung Pang,Huang Huan,Simple. Design of heart rate oximetry monitoring system based on Hongmeng system[J/OL]. Internet of Things Technology, 2025,3(15):1-4.

[4] Zhong Ming. Exploration of the Research on System Architecture and Application Development of Hongmeng[J]. Information and Computer (Theoretical Edition),2024,36(13):97-99.

[5] WANG Jianzhong,LIU Dan,DONG Minghua. Design and implementation of weather forecasting based on Hongmeng system[J]. Information and Computer(Theoretical Edition),2023,35(17):161-164+195.

[6] Luo Jiayang,Qu Yujia,Li Wenwen. The digital revolution of national fitness: value analysis, realistic dilemma and path choice[C]//International Bandy Federation,International Physical Fitness Association,Chinese Bandy Association.2024 Proceedings of the First International Digital Sports Science Conference (Upper). Sports Training Centre, Institute of Sports Science, General Administration of Sport of China; School of Education, Beijing Sport University; School of Physical Education, Kunming University of Technology;,2024:547-552.

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